

Week 1: The Evolution of Data

Lesson 1: The Evolution of Data

Data Everywhere

“You Are Creating Data Every Second”

Page 1: You Are Creating Data Right Now

Title: *Data Creation in Everyday Life*

Key Idea

- You are generating data this very moment.

Examples of Daily Data Creation

- Scrolling through your phone
- Taking photos
- Searching online
- Checking GPS location
- Wearing a smartwatch

Why This Matters

- Every interaction → part of the **global data ecosystem**
- Your actions feed into the **continuous stream of digital information**

Concept Reminder

- ◆ **Data is abstract**
 - ◆ **Data is constant**
 - ◆ **Data is connected to your daily actions**
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Page 2: Data Is Everywhere Around Us

Title: *Data in the Physical World*

Expanding the Idea

Data isn't only created by deliberate actions — it exists **around you**, woven into the environment.

Examples

- **Smart homes** monitoring electricity, temperature, appliances
- **Connected cars** sending real-time traffic and safety data
- **Smart city systems** tracking:
 - Public safety
 - Water usage
 - Traffic flow
 - Waste management
- Even **street lights** are part of data networks:
 - Adjusting brightness
 - Measuring pedestrian flow
 - Conserving energy

Understanding This

- We live inside a **continuous data ecosystem**
 - Countless sensors → constant information flow
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Page 3: The Scale of Data Today

Title: *2.5 Quintillion Bytes Per Day*

Important Number

- Humanity produces **2.5 quintillion bytes** of data *daily*.

Why This Feels Overwhelming

- The volume is beyond human capacity.
- To us, it might seem like **noise**, not information.

What We Need

- A way to **interpret**, **filter**, and **make sense** of the chaos.
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Page 4: How AI Makes Sense of Massive Data

Title: *AI as a Filter, Interpreter & Amplifier*

Role of AI

AI turns raw data into meaningful insights by:

- Identifying patterns
- Learning from large datasets
- Filtering irrelevant information
- Highlighting important connections

Analogy

🎵 **AI is like a master musician.**

- Untrained ears hear only noisy individual notes
- A master arranges them into a beautiful melody

Similarly

- Raw data = noise
- AI = transforms noise into meaning

Outcome

- Helps us understand our surroundings better
- Enables smarter decisions
- Converts complexity into clarity

Page 5: The Big Idea – Data + AI Shapes Our World

Title: *From Noise to Intelligence*

Combined Power

- Continuous data + powerful AI systems →
 - Predictive insights
 - Automation
 - Personalization
 - Smarter environments

Everyday Impact

- Accurate traffic routes
- Energy-efficient homes
- Personalized content recommendations

- Safer, faster, smarter cities

Final Thought

You're not just *using* the digital world —

✦ **You are shaping it, second by second, through the data you create.**
